EfficientNet-based Diabetic Retinopathy Classification Using Data Augmentation

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***Abstract*—Diabetic Retinopathy(DR) is an eye disease that leads to visual defects in people who have diabetes. Diabetes also causes the chance of arising other eye problems, including cataracts and glaucoma. Hence detection of symptoms associated with DR at early stages is essential. The DR grading process is complex due to tiny lesions, data inconsistency, and variation within the class. The solution for fine-grained DR grading is finding different features that affect the visual difference, such as Changes in blood vessel diameter, microaneurysms, soft exudates, hard exudates, and hemorrhages. However, small lesions are difficult to distinguish using convolutional neural networks (CNNs). The uneven distribution of DR data causes the model to focus on DR levels with many data, affecting the grading performance. The main aim of this paper is to explore the impact of various deep neural networks to improve scoring performance and address all the above challenges. Finally for improving the model performance, we introduce an image augmentation technique using the DDR dataset and compare the model's reaction. The results show that data augmentation can significantly improve classification performance compared to baselines.**

***Index Terms*—CNN, deep learning, DNN, Diabetic Retinopathy**

I. Introduction

Diabetic retinopathy is an eye-related disease that comes to people having long-term diabetes. This disease injures the tiny blood vessels of the retina, causing blood and fluid to leak out.DR presents asymptomatic/visual abnormalities in the early stages. However, it gradually reduces a person's vision and can even lead to complete blindness. In most developing and developed countries, it was the prime cause of sightlessness in people aged 25 to 55. However, regular and repeated annual retinal screening can reduce the risk of blindness in these patients by 95%. Therefore, early screening of the retina in diabetic patients is an important concern. An image of the retina with various symptoms is given in Fig.1. Visual information is acquired by the retina. The human eye can be overstimulated by eye diseases. In particular, retinal diseases can be identified in retinal fundus images. Changes in the diameter of the vessels, microaneurysms, exudate with lipid and protein deposits, and hemorrhage is the main symptoms of diabetic retinopathy. Based on the type and number of abnormal areas in the fundus images, it is possible to split the fundus images into five divisions of DR such that no DR, mild DR, moderate DR, severe DR, and proliferative DR. [1]. It can be seen from the images given in Fig 2. If it is kept in this state for a long time, certain retinal blood vessels will be obstructed, at last causing severe sight loss and even blindness. Advanced image processing techniques and efficient classifiers can detect these conditions from retinal fundus images to automatically diagnose and prevent blindness. With recent advances in computing hardware, it has become very easy to apply deep learning techniques that are beneficial to the biomedical field. Therefore, the use of deep learning improves DR diagnosis compared to traditional methods.

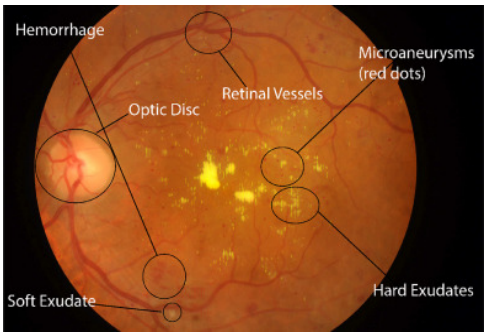

Fig. 1. Sample image of DR

In recent years, CNN has kept great success and has become widely used in computer science. CNN provides unique extraction and full-length classification and accomplished improvement in image classification [2], detection [3], semantic segmentation [4], and other tasks. X.Li and Wang [5] used a pre-trained CNN and apply transfer learning for DR representation. More clearly, they use a pre-trained CNN as a feature extractor and combine the output from the last fully connected layer with the support vector machine (SVM) to solve the scoring assignment. Y. Yang and Li [6] present a two-step CNN to find out lesions and DR levels in fundus images and obtained acceptable results. This mechanism processes DR ratings in two ways, which is more intricate than the single-step approach.

Even though the classification of Diabetic retinopathy using a convolutional neural network has been accomplished great comes out, its real-time clinical demand is even now difficult

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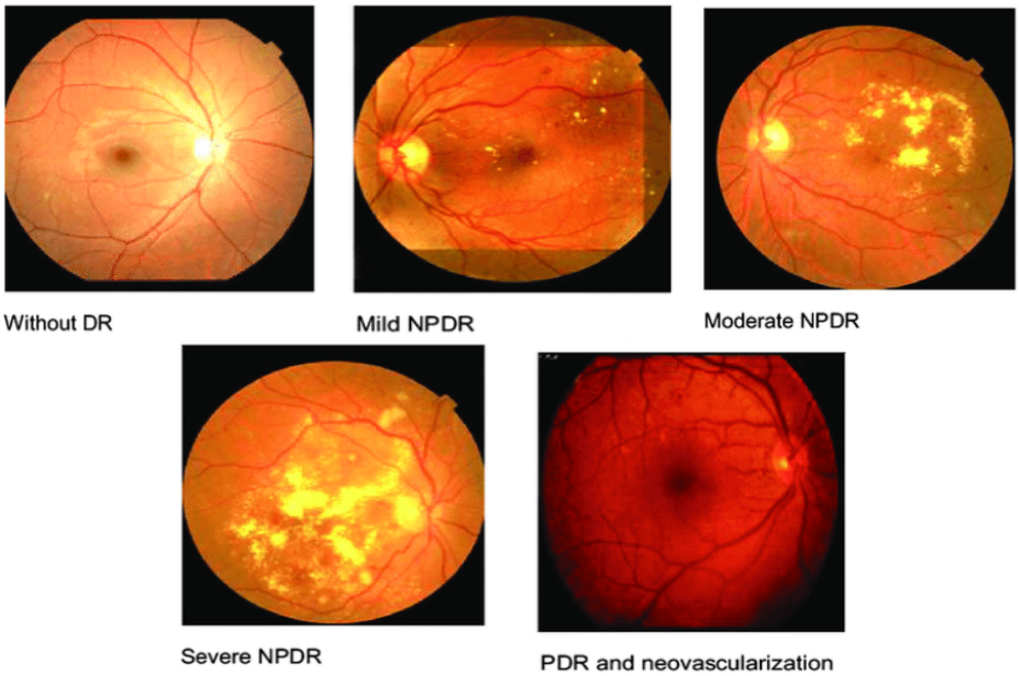

Fig. 2. Image of DR with varying levels of severity.

due to the complexity of the assignment. Foremost, as shown in Fig. 3 The distribution of Diabetic retinopathy samples among different stages is highly disproportioned since just a mini portion of Diabetic retinopathy samples have symptoms.

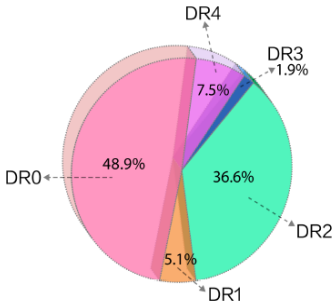

Fig. 3. The imbalanced data distribution of DDR Dataset

The disproportioned arrangement will cause the model to concentrate on DR stages with extra data (no DR indicated by DR0 and moderate DR indicated by DR2) and neglect those with a little amount of data (mild DR indicated by DR 1, severe DR indicated by DR 3, and proliferative DR indicated by DR 4), this will alter the model's generalization capability. Second, the colors and textures of the five different classes are similar, so the likelihood of confusion in the classification task is high. Third, as shown in Figure 2 certain lesions in the fundus image are of small size means that need only a few pixels to represent, and hence during the convolution process those few pixels will easily get ignored, which negatively affects the result of the Diabetic retinopathy representation. Motivated by the above challenges, we introduce two models 1) a five-layer CNN-based model and 2) an uninitialized EfficientNet-B3 model [7]. A brief sketch of the proposed models is described in Module II. The performance evaluation, including the detailed dataset description, is explained in Module III, and afterward, the result analysis is in Module IV. At last, this paper is concluded in Module V.

II. System Description

Convolutional neural networks are one of the most popular deep neural networks. A simple CNN consists of many layers, each of which transforms one volume of activations into another using a differentiable function. Convolutional, pooling, and dense layers are the three layers in CNN. In our work, we have a five-layer CNN model and another untrained EfficientNet-B3 deep CNN for feature extraction from fundus images. In the five-layer CNN-based model we have an 11x11 filter on the first layer and 7x7 filters on the remaining four layers. Larger filter sizes capture more spatial information because they consider larger pixel neighborhoods. The choice of filter size affects the types of features that can be detected. For example, a smaller filter size may be useful for detecting edges or corners, while a larger filter size may be better for detecting more complex patterns, such as textures or shapes. Filters are often shared between different regions of an image. This allows networks to learn translation-invariant features, meaning they detect the same feature no matter where it appears in the image. Filter size affects the degree of translation invariance, with larger filters providing more translation invariance. The input fundus image of resolution 227X227 is fed to a 2D convolution layer of filter size 32 and kernel size 11×11, followed by the ReLU activation layer

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and max-pooling layer. Again it passes through a series of convolution layers (filter size=64, 128, 256, 512, and kernel size=7) and max-pooling layers. The stride size of the max-pooling layer is 2. The proposed design of the five-layer CNN model is shown in Table I

The second model is untrained EfficientNet-B3.We trained the algorithm with the same neural network architecture as in the original study of Mingxing Tan and Quoc V. Le [7]. EfficientNet-B3 is one of the many variants of the EfficientNet architecture designed to achieve high accuracy on various image classification metrics while keeping model size and computational cost low. They put forward an advanced scaling formula that equally scales every single depth or width or resolution dimension using an effortless but effective compound coefficient mechanism. They laid a new foundation and went on to achieve a family of models called EfficientNets that attain considerable accuracy and efficiency than prior neural networks.EfficientNet-B3 is one of the models in the EfficientNet family designed to increase computational efficiency while performing the latest technology image classification. The designed architecture of the model EfficientNet-B3 is given in Fig.4. The EfficientNet-B3 architecture can be divided into three main components: the stem, the intermediate blocks, and the top.

The stem is the first part of the network and is responsible for processing the input image. In EfficientNet-B3, the stem comprises a series of convolutional layers, subsequently the max pooling layer. The resultant output from the stem is then given to the intermediate blocks. The intermediate blocks are a series of repeating blocks that are responsible for extracting features from the input image. Each block consists of several layers, including convolutional layers, batch normalization, and activation functions. EfficientNet-B3 uses a total of 7 intermediate blocks. The top is the final part of the network and is responsible for producing the output classification. In EfficientNet-B3, the top consists of a flattened layer, it will convert the output of the EfficientNetB3 layer (which is a tensor with a shape of (None, 9, 9, 1536) for EfficientNetB3) into a one-dimensional tensor with a shape of (None, 9 * 9 * 1536) and a softmax activation function to produce the final probability distribution over the classes. One notable feature of the EfficientNet family of models is that they use a technique called compound scaling to efficiently increase the number of parameters and improve performance. This includes scaling the depth/ width/resolution of the network in a coordinated way, using a set of scaling coefficients that are determined empirically. In the example of EfficientNet-B3, the network has about 12 million parameters and achieves the most advanced performance of many image classification models. The architecture of the EfficientNet-B3 model is shown in Table II

TABLE I
FIVE LAYER-CNN BASED MODEL

No	Layer(Type)	Output shape
1	conv2d (Conv2D)	(None, 217, 217, 32)
2	maxpooling2d (MaxPooling2D)	(None, 108, 108, 32)
3	conv2d1(Conv2D)	(None, 102, 102, 64)
4	maxpooling2d1 (Max-Pooling2D)	(None, 51, 51, 64)
5	conv2d2(Conv2D)	(None, 45, 45, 128)
6	maxpooling2d2 (Max-Pooling2D)	(None, 22, 22, 128)
7	conv2d3(Conv2D)	(None, 16, 16, 256)
8	maxpooling2d3 (Max-Pooling2D)	(None, 8, 8, 256)
9	conv2d4(Conv2D)	(None, 2, 2, 512)
10	maxpooling2d4 (Max-Pooling2D)	(None, 1, 1, 512)
11	flatten(Flatten)	(None, 512)
12	dense (Dense)	(None, 512)
13	dropout (Dropout)	(None, 512)
14	dense1 (Dense)	(None, 256)
15	dense2 (Dense)	(None, 5)

TABLE II
MODEL EFFICIENTNET-B3

No	Layer(Type)	Output shape
1	efficientnet-b3	(None, None, None, 1536)
2	flattent(Flatten)	(None, 98304)
3	dense(Dense)	(None, 5)

III. PERFORMANCE EVALUATION

A. Dataset

The DDR database holds 13,673 fundus photographs, counting 6,835 training photographs, 2,733 validation photograph, and 4,105 test photographs. The images were classified by seven trained raters into six categories according to the International classification of diabetic retinopathy. Second-rate photographs that do not have a clear view are considered unrated. So the six classes are no DR, mild DR, moderate DR, severe DR, proliferative, and ungradable DR. No DR means no evidence of retinal vascular disease. Indications $\geq$ A microaneurysm with no other findings is considered a mild DR. Images containing $\geq$ 20 intraretinal hemorrhage in quadrants 1-3 were classified as moderate DR. Four quadrants of $\geq$ 20 hemorrhages show severe DR. In the experiment, our focus is just on the five-class grading assignment for improving DR notations; i.e. we do not use those images that are in the unevaluable classes. Therefore, the training, validation, and test images are 11220, , 1244 and 125 respectively.

B. Performance metrics

We evaluate the efficacy of the proposed system using precision, recall, specificity, F1 score, and accuracy.

- Precision: It is the amount of information that is conveyed by a value.

$$Precision = \frac{TP}{(TP + FP)} \tag{1}$$

- Recall: Otherwise known as true positive rate (TPR), this rate is the percentage of data samples that a machine

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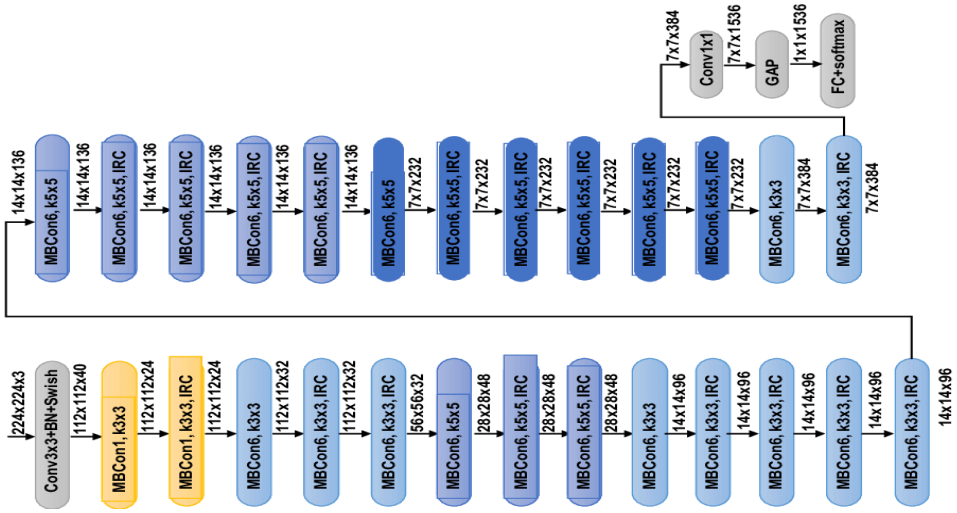

Fig. 4. Architecture of EfficientNet-B3

learning model identifies as classes of interest ("good classes") out of all samples of that class.

$$Recall = \frac{TP}{(TP + FN)} \tag{2}$$

- Specificity: Specificity is the proportion of true negatives that the model accurately predicts.

$$Specificity = \frac{TN}{(TN + FP)} \tag{3}$$

- F1 score: F1 score is defined as the mean of precision and recall.

$$F1score = 2 * \frac{precision * recall}{precision + recall} \tag{4}$$

- Accuracy: Accuracy is usually a metric that describes how well the system performs in every class. This applies when all the categories have equal significance. This can be computed as the proportion of the number of correct guesses to the total number of guesses.

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \tag{5}$$

C. Experimental Framework

Four systems were experimented with in our proposed work. First, we experimented with CNN without augmentation. We then experimented with the same network but used augmented data. Next, we designed an untrained efficientNet-B3 and experimented without augmentation. Finally, with untrained efficientNet-B3, use augmented data. We use Relu as the activation function in the five-layer CNN model. Compared with Sigmoid and TanH, ReLu is currently the best and most efficient function, because all the negatives such as the vanishing gradient problem in this activation function are eliminated, which makes this activation function more powerful compared to other activation functions. The optimizer used here is Adam. The results of the Adam optimizer are generally better than any other optimization with faster computation times and fewer parameters.

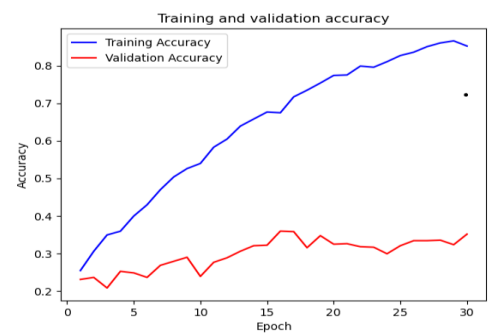

Fig. 5. Training and validation accuracy curve

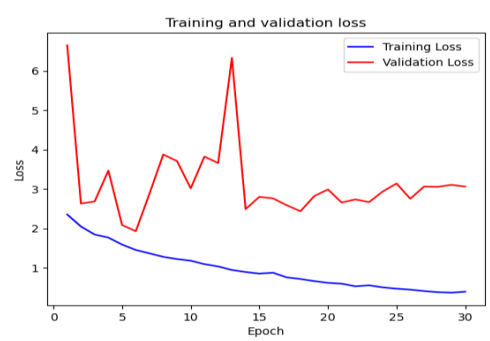

Fig. 6. Training and validation loss curve

The EfficientNet architecture uses Swish at all convolutional and all link layers. Swish is a smooth, non-monotonic activation function explained in such a way that:

$$Swish(x) = x * sigmoid(\beta * x) \tag{6}$$

where β is a hyperparameter that controls the smoothness of the functionality. The Swish activation function performs

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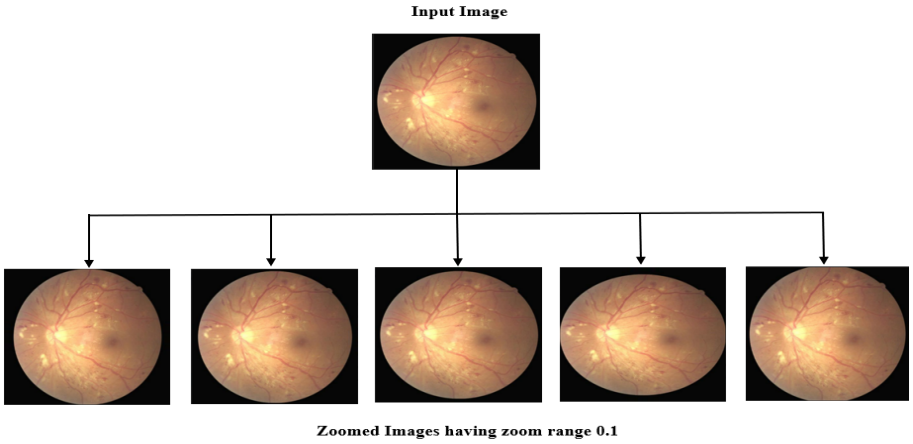

Fig. 7. Augmented images

better than other popular activation functions such as ReLU, LeakyReLU, and ELU in various tasks. In addition to the Swish activation function, the EfficientNet also uses other techniques such as the mobile inverted bottleneck convolution and compound scaling to achieve a high level of efficiency and accuracy. Softmax activation is used in both models for classification. The models are iterated over 30 epochs with a batch size of 100 for a five-layer CNN-based model and a batch size of 4 for the untrained EfficientNet-B3 model. The categorical cross-entropy loss function and a checkpoint that monitors validation accuracy are also employed for experiments.Fig.5.indicates the training and validation accuracy curve obtained for different epochs for the model EfficientNet-B3. Similarly, Fig.6.shows the loss values obtained for different epochs.

D. Data Augmentation

Image data augmentation is one of the popular types of data augmentation and involves transforming images in data and is used to supplement training data to improve model performance. We have common image transformations such as translation, stretch, rotation, scroll, and color scaling for data augmentation. But in our work, we only focus on augmentation by Zoom. Because we thought there might be an opportunity to get more features by doing so. This step solves a major problem of overfitting caused by unbalanced data distribution.Fig.7. showing the images obtained after augmentation.

IV. Results and Analysis

We tested our model on data from 125 previously unseen fundus images. We experimented with four phases: CNN with and without augmentation and EfficientNetB3 with and without augmentation. The outcomes are tabulated in Table III. The total accuracy reported is 75%, 93%, 79%, and 94%, for five-layer CNN with no augment, five-layer CNN with augment, EfficientNetB3 without augment, and EfficientNetB3 with augment, respectively.

The distribution of Diabetic retinopathy samples among different stages is highly disproportioned since just a mini portion of DR samples have symptoms. The disproportioned arrangement will cause the model to concentrate on DR stages with extra data (no DR and moderate DR) and neglect those with a little amount of data (mild DR, severe DR, and proliferative DR) will alter the model's generalization capability. This is why we get bad results even if the model is good. In our experiments, the best-performing untrained EfficientNet-B3 model outperforms all other architectures with an accuracy of 94%.

TABLE III
MULTI-CLASS DETECTION RESULTS

Model	Precision	Recall	Specificity	F1score	Accuracy
CNN without Augment	0.27	0.40	0.85	0.30	0.75
CNN with Augment	0.83	0.83	0.96	0.81	0.93
EfficientNet-B3 without augment	0.57	0.47	0.86	0.43	0.79
EfficientNnet-B3 with augment	0.87	0.86	0.96	0.86	0.94

TABLE IV
MULTI-CLASS DETECTION RESULTS OF FIVE-LAYER CNN WITH AUGMENTATION.

Types	Precision	Recall	Specificity	F1score	Accuracy
None	0.79	0.92	0.94	0.85	0.94
Mild	0.86	1.00	0.96	0.93	0.97
Moderate	0.83	0.40	0.98	0.54	0.86
Severe	0.91	0.84	0.98	0.88	0.95
Proliferative	0.78	1.00	0.93	0.88	0.94

When we experiment with the five-layer CNN, it correctly predicts all 25 samples for the two classes and also gives good results for the remaining classes except grade two. Fig. 8 shows the confusion matrixes of the EfficientNet-B3 model with and without data augmentation. From the confusion matrix, it is clear that the EfficientNet-B3 model

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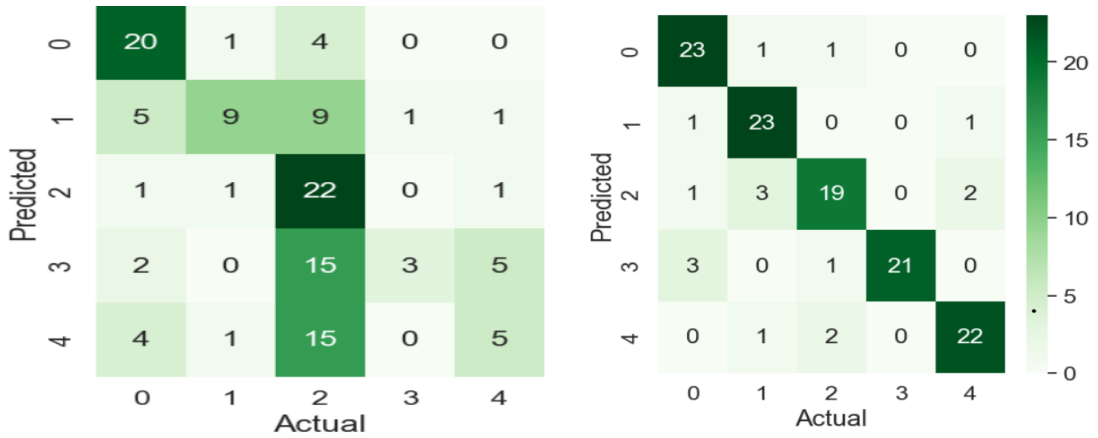

Fig. 8. Confusion Matrix of EfficientNetB3 (a) without augmentation (b) with augmentation

TABLE V
MULTI-CLASS DETECTION RESULTS OF EFFICIENTNETB3 WITH AUGMENTATION.

Types	Precision	Recall	Specificity	F1score	Accuracy
None	**0.82**	**0.92**	**0.95**	**0.87**	**0.94**
Mild	**0.82**	**0.92**	**0.95**	**0.87**	**0.94**
Moderate	**0.83**	**0.76**	**0.96**	**0.79**	**0.92**
Severe	**1.00**	**0.84**	**1.00**	**0.91**	**0.97**
Proliferative	**0.88**	**0.88**	**0.97**	**0.88**	**0.95**

with data augmentation outperforms the other scheme without data augmentation. Results in Table III clearly show that introducing augmentation on our models creates distinguishable advances in the grading task, particularly when measuring performance in terms of specificity and test accuracy. The class-wise performance of the models with augmentation can be examined from Table IV and Table V. The accuracy for the moderate class is comparatively low for the five-layer CNN model. But it is improved significantly in the EfficientNet-B3 model.

While the performance is better, even now there is scope for improvement. First, the whole system is trained only with image-level tracking, hence there is still difficult to detect small lesions. Second, in terms of hospital use, our model can issue competitive results, but not the type of DR disease useful for DR screening, such as soft exudates, hard exudates, microaneurysms, and hemorrhages. Therefore, it should be mentioned in future work. Moreover, we will consider using bottleneck features and attention mechanisms to improve deployment.

V. CONCLUSION

Ophthalmologists can identify Diabetic retinopathy from the presence of lesions associated with abnormal blood vessels. When this method works, it has a high resource demand. In this study, we put two models, a five-layer CNN model, and an untrained EfficientNet-B3. Our future focus is to use Bottleneck features for deployment. Therefore, we can construct a further efficient model that can indicate the type of lesion and can provide a rank score.

REFERENCES

[1] C. P. Wilkinson et al., “Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales,” Ophthalmology, vol. 110, no. 9, pp. 1677-1682, 2003.
[2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,“ in CVPR, Las Vegas, NV, USA, 2016, pp. 770-778.
[3] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in NIPS, Montreal, Canada, 2015, pp. 91-99.
[4] K. He, G. Gkioxari, P. Dollar, and R. Girshick, “Mask R-CNN,” in ICCV, Venice, Italy, 2017, pp. 2980-2988.
[5] X. Li, T. Pang, B. Xiong, W. Liu, P. Liang, and T. Wang, “Convolutional neural networks based transfer learning for diabetic retinopathy fundus image classification,” in CISP-BMEI, Shanghai, China, 2017, pp. 1-11.
[6] Y. Yang, T. Li, W. Li, H. Wu, W. Fan, and W. Zhang, “Lesion detection and grading of diabetic retinopathy via two-stage deep convolutional neural networks,” in MICCAI, Quebec City, QC, Canada, 2017, pp. 533- 540.
[7] Mingxing Tan, and Quoc V. Le, ”EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks, ”Proceedings of the 36th International Conference on Machine Learning, Long Beach, California, PMLR 97, 2019.
[8] M. Saini and S. Susan ”Diabetic retinopathy screening using deep learning for multi-class imbalanced datasets, ”Computers in Biology and Medicine 149 (2022) 105989
[9] Haomiao Liu, Lu Teng, Linhua Fan, Yabin Sun, and Huiying Li “A new ultra-wide-field fundus dataset to diabetic retinopathy grading using hybrid preprocessing methods,” Computers in Biology and Medicine 157 (2023) 106750
[10] Along He, Tao Li, Ning Li, Kai Wang, and Huazhu Fu, “CABNet: Category Attention Block for Imbalanced Diabetic Retinopathy Grading,” IEEE Transactions on Medical Imaging,2017.
[11] Ziyuan Zhao, Kerui Zhang, Xuejie Haoz, Jing Tiany, Matthew Chin Heng Chuay, Li Chen, and Xin Xu, “BIRA-NET: BILINEAR ATTENTION NET FOR DIABETIC RETINOPATHY GRADING,” Published in the IEEE 2019 International Conference on Image Processing (ICIP 2019)
[12] Yehui Yang, Fangxin Shang, Binghong Wu, Dalu Yang, Lei Wang, Yanwu Xu, Wensheng Zhang and Tianzhu Zhang, “Robust Collaborative Learning of Patch-level and Image-level Annotations for Diabetic Retinopathy Grading from Fundus Image”
[13] Adrian Galdran1, José Dolz, Hadi Chakor, Hervé Lombaert, and Ismail Ben Ayed2, “COST-SENSITIVE REGULARIZATION FOR DIABETIC RETINOPATHY GRADING FROM EYE FUNDUS IMAGES,” Accepted at MICCAI 2020 - OCTOBER 2, 2020
[14] Yuhao Niu, Lin Gu, Yitian Zhao, and Feng Lu, “Explainable Diabetic Retinopathy Detection and Retinal Image Generation,” Generic Colorized Journal, 2017.
[15] Yijin Huang, Junyan Lyu, Pujin Cheng, Roger Tam, and Xiaoying Tang ”SSiT: Saliency-guided Self-supervised Image Transformer for Diabetic Retinopathy Grading,” IEEE Transactions on Medical Imaging, 2022.

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